



KTH Computer Science
and Communication

Exam in DD2421 Machine Learning 2024-10-24, kl 8.00 – 12.00

Aids allowed: *calculator, language dictionary.*

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$54 < p \leq 60 \rightarrow A$$

$$48 < p \leq 54 \rightarrow B$$

$$42 < p \leq 48 \rightarrow C$$

$$36 < p \leq 42 \rightarrow D$$

$$29 < p \leq 36 \rightarrow E \text{ (A pass is guaranteed with the required points for 'E'.)}$$

$$0 \leq p \leq 29 \rightarrow F$$

This exam consists of sections **A**, **B**, and **C**.

NB: use different papers (answer sheets) for different sections A, B, and C.

This means you shall start a new page after section A, and again after section B.

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology

(4p)

For each term (a–d) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

- | | |
|-------------------------------|---|
| | 1) Robust method to fit a model to data with outliers |
| | 2) An ensemble approach by generating k different models |
| a) k -means | 3) Method for estimating the mean of k observations |
| b) RANSAC | 4) Clustering method based on centroids |
| c) k -fold cross validation | 5) An efficient method to find the optimal trade-off |
| d) k -nearest neighbour | 6) Class prediction by a majority vote |
| | 7) Strategy to completely search a space while causing mess |
| | 8) A technique for assessing a model while exploiting available data for training and testing |

A-2 Nearest Neighbor, Classification

(4p)

Suppose that we take a data set, divide it into two parts of equal size, Part I and Part II. We try out two different classification procedures, by using Part I and Part II as our training set and test set, respectively. That is, we use half of the data for training, and the remaining half for testing.

- a) First we use 1-Nearest Neighbor rule (1-NN) and get an average error rate (averaged over both test and training data sets) of 8%. What was the error rate with 1-nearest neighbor on the test set? Briefly reason the answer. (1p)
- b) Next we use the Adaboost Algorithm and get an error rate of 10% on the training data. We also get the average error rate (averaged over both test and training data sets) of 12%. What was the error rate with the Adaboost Algorithm on the test set? (1p)
- c) Now, we swap the roles of Part I and Part II, and repeat the same experiments. On the test set (Part I), we get an error rate of 12% with both 1-NN and the Adaboost Algorithm. Based on all these results, by the cross-validation, indicate the method which we should prefer to use for classification of new observations, with a simple reasoning. (2p)

A-3 Entropy and Decision Trees/Forests

(5p)

- a) Consider a single toss of *skewed* coin (it is likely to show one side more than the other side). Regarding the uncertainty of the outcome {head, tail}, which one of the following is correct?

- i. the entropy is greater than one bit.
- ii. the entropy is equal to one bit.
- iii. the entropy is smaller than one bit.

Simply indicate your choice. (1p)

- b) Indicate any valid ones as the basic strategy for selecting a question (attribute) at each node in decision trees.

- i. To minimize the expected reduction of the entropy.
- ii. To minimize the expected reduction of gini impurity.
- iii. To maximize the expected reduction of the entropy.

Simply indicate your choice. (1p)

- c) Draw a decision tree of *depth two* that is consistent with the training data in the table. $a_1, \dots, a_4$ are attributes of the training data. (It is not required to use information gain to solve the task.) (2p)

a_1	a_2	a_3	a_4	$class$
0	0	0	0	-
0	0	1	1	+
0	1	0	1	+
0	1	1	0	-
1	0	0	0	+
1	0	1	1	+
1	1	1	0	-
1	1	0	1	-

- d) Suppose we have trained a Decision Forest using five bootstrapped samples from a data set containing three classes, {Red, Yellow, Green}. We then applied the model to a specific test input, $\mathbf{x}$, and observed five estimates of class distributions from the five trees, respectively:

$$\begin{aligned} \{P_1(\text{Red}|\mathbf{x}), P_1(\text{Yellow}|\mathbf{x}), P_1(\text{Green}|\mathbf{x})\} &= \{0.1, 0.5, 0.4\} \\ \{P_2(\text{Red}|\mathbf{x}), P_2(\text{Yellow}|\mathbf{x}), P_2(\text{Green}|\mathbf{x})\} &= \{0.2, 0.3, 0.5\} \\ \{P_3(\text{Red}|\mathbf{x}), P_3(\text{Yellow}|\mathbf{x}), P_3(\text{Green}|\mathbf{x})\} &= \{0.3, 0.6, 0.1\} \\ \{P_4(\text{Red}|\mathbf{x}), P_4(\text{Yellow}|\mathbf{x}), P_4(\text{Green}|\mathbf{x})\} &= \{0.2, 0.4, 0.4\} \\ \{P_5(\text{Red}|\mathbf{x}), P_5(\text{Yellow}|\mathbf{x}), P_5(\text{Green}|\mathbf{x})\} &= \{0.2, 0.3, 0.5\} \end{aligned}$$

Let us consider the average probability for combining these results into a single class prediction. What will the final class prediction be? *Motivate your answer by short phrases.* (1p)

A-4 Regression with regularization

(5p)

- a) In regression, regularization is a process of introducing additional term, so-called shrinkage penalty. Which ones of the three methods includes the additional term.
- Logistic regression.
 - Ridge regression.
 - k -NN regression.

Simply indicate your choice. (1p)

- b) For a set of N training samples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, each consisting of input vector $\mathbf{x}$ and output y , suppose we estimate the regression coefficients $\mathbf{w}^\top (\in \mathbf{R}^d) = \{w_1, \dots, w_d\}$ in a linear regression model by minimizing

$$\sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 \quad \text{subject to} \quad \sum_{i=1}^d |w_i| \leq s$$

for a particular value of s . As we *increase* s from 0, the training error and test errors (residual sum of squares, RSS) are believed to change. Indicate which of {i,ii,iii,iv,v} is correct about the *test* error, and *justify your answer*. (2p)

- Remain constant.
 - Steadily decrease.
 - Steadily increase.
 - Decrease initially, and then eventually start increasing in a U shape.
 - Increase initially, and then eventually start decreasing in an inverted U shape.
- c) As we *increase* s from 0, what happens to the variance. Choose the answer from the five alternatives above, and *justify your answer*. (1p)
- d) The method described in **b)** is called LASSO and known to yield *sparse models*. Briefly explain what property of it enables the sparsity in a short sentence. (1p)

A-5 Dimensionality Reduction, Subspace Methods

(4p)

We consider to solve a K -class classification problem with the Subspace Method and for that we compute a subspace $\mathcal{L}^{(j)}$ ($j = 1, \dots, K$) using training data for each class, respectively. That is, given a set of feature vectors (as training data) which belong to a specific class C (i.e. with an identical class label), we perform PCA on them and generate an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ which spans a p -dimensional subspace, $\mathcal{L}$, as the outcome.

Provide an answer to the following questions.

- Given that we compute $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ as eigenvectors of the auto-correlation matrix Q based on the training data, how should we choose the eigenvectors in relation to the corresponding eigenvalues of Q ? (1p)
- We have a new input vector $\mathbf{x}$ whose class is unknown, and consider its projection length on each $\mathcal{L}^{(j)}$. Describe how the projection length is represented, using a simple formula. (2p)
- Briefly explain the way to determine the class to which vector $\mathbf{x}$ should belong using the projection lengths. (1p)

B Graded problems

Potential inquiries to be addressed to Olov Andersson.

B-1 Probability Theory

(4p)

You are training a binary classifier to detect whether an email is spam or not. The following probabilities are given:

- $P(\text{Spam}) = 0.3$ – The probability that a randomly selected email is spam.
- $P(\text{Offer} \mid \text{Spam}) = 0.8$ – The conditional probability that an email contains the word "Offer" given that it is spam.
- $P(\text{Offer} \mid \neg \text{Spam}) = 0.2$ – The conditional probability that the email contains the word "Offer" given that it is NOT spam.
- $P(\text{Urgent} \mid \text{Spam}) = 0.6$ – The conditional probability that the mail contains the word "Urgent" given that it is spam.
- $P(\text{Urgent} \mid \neg \text{Spam}) = 0.1$ – The conditional probability that the mail contains the word "Urgent" given that it is NOT spam.

Using Bayes' Theorem, answer the following:

- a) What is the probability that an email containing the word "Offer" is spam? (**Calculate** $P(\text{Spam} \mid \text{Offer})$) (2p)
- b) Suppose the classifier also checks for the word "Urgent" in emails. Now **calculate** $P(\text{Spam} \mid \text{Offer}, \text{Urgent})$. You may assume that the random variables Offer and Urgent are independent. (2p)

B-2 Maximum Likelihood Estimation

(5p)

Suppose you are building a machine learning model to predict the probability of a coin landing heads up. You think that the coin may have a bias, where the probability of heads θ is in the range $0 \leq \theta \leq 1$. You perform an experiment where you flip the coin 10 times, and it lands heads 7 times.

- a) **Write** the likelihood function $L(\theta)$ for the observed data above. Hint: The probability of a coin landing heads is θ , which means that the probability of tails is... (2p)
- b) **Derive** the maximum likelihood estimate (MLE) for the parameter θ . Show your calculations (2p)
- c) What is the relation between the MLE and the true value of the parameter? (1p)

B-3 Maximum a Posteriori Estimation

(3p)

Consider a simple linear regression model given by:

$$y_i = wx_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

where y_i is the observed output, x_i is the input feature, w is the parameter to be estimated, and ϵ_i is Gaussian noise with mean 0 and variance σ^2 .

Suppose you have the following prior distribution for the parameter w :

$$p(w) \sim \mathcal{N}(0, \tau^2)$$

where τ^2 is the prior variance. You are given N observations $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$.
Hint: The pdf of a variable x with a Gaussian (Normal) distribution has the form

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \quad (1)$$

- a) **Derive the Maximum A Posteriori (MAP)** estimate for the parameter w . (2p)
- b) **Compare** the MAP estimator for w to the Maximum Likelihood Estimator (MLE) as $\tau \rightarrow \infty$. (1p)

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

C-1 Multiple-Choice: Support Vector Machine

(1p)

Do not justify your answer. Instead, select exactly **one** option of (1.), (2.), or (3.).

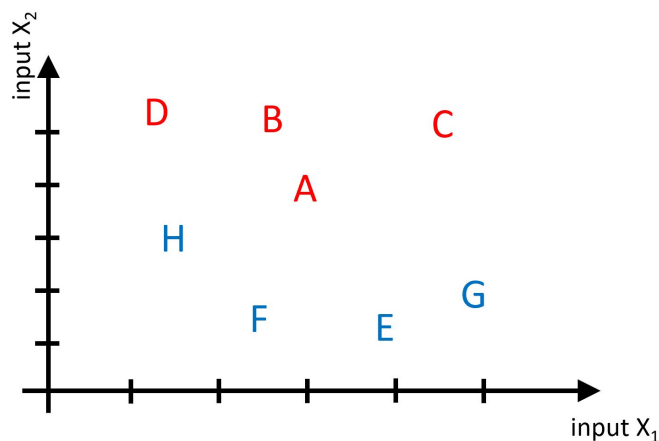
Complete the following sentence: Out of all hyperplanes which solve a classification problem, the one with widest margin will probably ...

1. ... compute fastest.
2. ... have the smallest number of parameters.
3. ... generalize best.

C-2 Support Vector Classification

(3p)

The following diagram shows a small data set consisting of four RED samples (A, B, C, D) and four BLUE samples (E, F, G, H). This data set can be linearly separated.



- a) We use a linear support vector machine (SVM) without kernel function to correctly separate the BLUE and the RED class. Which of the data points (A-H) can be removed for training without changing the resulting SVM decision boundary? Name the point(s) that can be removed, and justify with **KEYWORDS** (or very short sentences). (2p)
- b) Assume someone suggests using a non-linear kernel for the SVM classification of the above data set (A-H). Give one argument **AGAINST** using non-linear SVM classification for such a data set. **USE KEYWORDS!** (1p)

C-3 Multiple-Choice: Artificial Neural Networks

(1p)

Do not justify your answer. Instead, select exactly **one** option of (1.), (2.), or (3.).
Error-Backpropagation-Training for Neuronal Networks requires the following:

1. Gaussian distributed training data and a fast computer.
2. labeled training data and differentiable neuron activation functions.
3. a single input and a single output to the network.

C-4 Artificial Neural Networks

(3p)

Consider the training data in the table, where + means a positive sample and – a negative.

- a) What is the minimum number of *layers* needed for an artificial neural network to correctly classify all these points? Motivate your answer **IN KEYWORDS**. (2p)
- b) What should the number of input and output nodes be in this network (assume you want a minimal working network)? Motivate your answer **IN KEYWORDS**. (1p)

x_1	x_2	Class
4	4	+
4	-2	+
2	0	–
0	2	–
-3	-3	+
-3	4	+